Year 11 Foundation

REVISION PACK

Progress Check 3

ANSWERS

Units in this pack

- Number 9
- Algebra 8

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Number 9 · Percentages of amounts (Lesson 6.1)

Q	Working	Answer
a percentage of an amount		
1.1	$15\% \text{ of } 80 = \frac{15}{100} \times 80$ $= 12$	**12 KD**
1.2	$65\% \text{ of } 360 = \frac{65}{100} \times 360$ $= 234$	**234 KD**
1.3	$12\% \text{ of } 650 = \frac{12}{100} \times 650$ $= 78$	**78 KD**
1.4	$25\% \text{ of } 180 = \frac{25}{100} \times 180$ $= 45$	**45 KD**
1.5	$5\% \text{ of } 340 = \frac{5}{100} \times 340$ $= 17$	**17 KD**
1.6	$75\% \text{ of } 96 = \frac{75}{100} \times 96$ $= 72$	**72 KD**
writing one number as a percentage of another		
1.7	$\frac{18}{25} \times 100$ $= 72\%$	**72%**
1.8	$\frac{39}{60} \times 100$ $= 65\%$	**65%**
1.9	$\frac{63}{84} \times 100$ $= 75\%$	**75%**

1.10	$\frac{27}{30} \times 100$ $= 90\%$	**90%**
1.11	$\frac{21}{35} \times 100$ $= 60\%$	**60%**
1.12	$\frac{56}{80} \times 100$ $= 70\%$	**70%**
a percentage of the rest		
1.13	$\text{rest} = 1000 \times 0.8 = 800$ $\text{rent} = 800 \times 0.3 = 240$	**240 KD**
1.14	$\text{rest} = 1500 \times 0.9 = 1350$ $\text{rent} = 1350 \times 0.4 = 540$	**540 KD**
1.15	$\text{rest} = 800 \times 0.75 = 600$ $\text{food} = 600 \times 0.5 = 300$	**300 KD**
1.16	$\text{rest} = 2000 \times 0.8 = 1600$ $\text{food} = 1600 \times 0.25 = 400$	**400 KD**
1.17	$\text{rest} = 1200 \times 0.75 = 900$ $\text{rent} = 900 \times 0.2 = 180$	**180 KD**
1.18	$\text{rest} = 900 \times 0.9 = 810$ $\text{food} = 810 \times 0.5 = 405$	**405 KD**

Number 9 · Percentage increase, decrease and profit (Lesson 6.2)

Q	Working	Answer
increasing an amount		
2.1	$\text{multiplier} = 1.1$ $40 \times 1.1 = 44$	**44 KD**
2.2	$\text{multiplier} = 1.14$ $75 \times 1.14 = 85.5$	**85.5 KD**

2.3	$\text{multiplier} = 1.08$ $250 \times 1.08 = 270$	**270 KD**
2.4	$\text{multiplier} = 1.05$ $120 \times 1.05 = 126$	**126 KD**
2.5	$\text{multiplier} = 1.25$ $64 \times 1.25 = 80$	**80 KD**
2.6	$\text{multiplier} = 1.3$ $90 \times 1.3 = 117$	**117 KD**
decreasing an amount		
2.7	$\text{multiplier} = 0.9$ $3000 \times 0.9 = 2700$	**2700 KD**
2.8	$\text{multiplier} = 0.85$ $7200 \times 0.85 = 6120$	**6120 KD**
2.9	$\text{multiplier} = 0.75$ $480 \times 0.75 = 360$	**360 KD**
2.10	$\text{multiplier} = 0.8$ $350 \times 0.8 = 280$	**280 KD**
2.11	$\text{multiplier} = 0.95$ $9000 \times 0.95 = 8550$	**8550 KD**
2.12	$\text{multiplier} = 0.85$ $640 \times 0.85 = 544$	**544 KD**
percentage profit		
2.13	$\text{profit} = 60 - 50 = 10$ $\frac{10}{50} \times 100 = 20\%$	**20%**
2.14	$\text{profit} = 46 - 40 = 6$ $\frac{6}{40} \times 100 = 15\%$	**15%**
2.15	$\text{profit} = 100 - 80 = 20$ $\frac{20}{80} \times 100 = 25\%$	**25%**

2.16	$\text{profit} = 78 - 60 = 18$ $\frac{18}{60} \times 100 = 30\%$	**30%**
2.17	$\text{profit} = 220 - 200 = 20$ $\frac{20}{200} \times 100 = 10\%$	**10%**
2.18	$\text{profit} = 20 - 16 = 4$ $\frac{4}{16} \times 100 = 25\%$	**25%**

Number 9 · Reverse percentages (Lesson 6.4)

Q	Working	Answer
price before an increase		
3.1	$120\% = 72$ $\text{original} = 72 \div 1.2 = 60$	**60 KD**
3.2	$110\% = 55$ $\text{original} = 55 \div 1.1 = 50$	**50 KD**
3.3	$130\% = 130$ $\text{original} = 130 \div 1.3 = 100$	**100 KD**
3.4	$105\% = 84$ $\text{original} = 84 \div 1.05 = 80$	**80 KD**
3.5	$125\% = 150$ $\text{original} = 150 \div 1.25 = 120$	**120 KD**
3.6	$115\% = 46$ $\text{original} = 46 \div 1.15 = 40$	**40 KD**
price before a reduction		
3.7	$75\% = 36$ $\text{original} = 36 \div 0.75 = 48$	**48 KD**
3.8	$65\% = 52$ $\text{original} = 52 \div 0.65 = 80$	**80 KD**

3.9	$90\% = 63$ $\text{original} = 63 \div 0.9 = 70$	**70 KD**
3.10	$90\% = 45$ $\text{original} = 45 \div 0.9 = 50$	**50 KD**
3.11	$80\% = 56$ $\text{original} = 56 \div 0.8 = 70$	**70 KD**
3.12	$60\% = 120$ $\text{original} = 120 \div 0.6 = 200$	**200 KD**

Algebra 8 · Simultaneous equations (Lesson 12.1)

Q	Working	Answer
multiplying both equations		
4.1	$3x + 2y = 14$ $2x + 3y = 11$ $\times 3: \ 9x + 6y = 42$ $\times 2: \ 4x + 6y = 22$ $\text{subtract}: \ 5x = 20$ $x = 4$ $3 \times 4 + 2y = 14$ $y = 1$	$x = 4, y = 1$
4.2	$5x + 2y = 9$ $3x + 4y = 11$ $\times 4: \ 20x + 8y = 36$ $\times 2: \ 6x + 8y = 22$ $\text{subtract}: \ 14x = 14$ $x = 1$ $5 \times 1 + 2y = 9$ $y = 2$	$x = 1, y = 2$

4.3	$2x + 5y = 11$ $3x + 2y = 11$ $\times 2: \ 4x + 10y = 22$ $\times 5: \ 15x + 10y = 55$ $\text{subtract}: \ 11x = 33$ $x = 3$ $2 \times 3 + 5y = 11$ $y = 1$	$x = 3, y = 1$
4.4	$4x + 3y = 18$ $3x + 2y = 13$ $\times 2: \ 8x + 6y = 36$ $\times 3: \ 9x + 6y = 39$ $\text{subtract}: \ x = 3$ $4 \times 3 + 3y = 18$ $y = 2$	$x = 3, y = 2$
4.5	$2x + 3y = 14$ $5x + 4y = 21$ $\times 4: \ 8x + 12y = 56$ $\times 3: \ 15x + 12y = 63$ $\text{subtract}: \ 7x = 7$ $x = 1$ $2 \times 1 + 3y = 14$ $y = 4$	$x = 1, y = 4$
4.6	$3x + 4y = 10$ $2x + 5y = 9$ $\times 5: \ 15x + 20y = 50$ $\times 4: \ 8x + 20y = 36$ $\text{subtract}: \ 7x = 14$ $x = 2$ $3 \times 2 + 4y = 10$ $y = 1$	$x = 2, y = 1$
different signs		

4.7	$3x + 2y = 9$ $4x - y = 1$ $\times 2: \ 8x - 2y = 2$ $\text{add}: \ 11x = 11$ $x = 1$ $3 \times 1 + 2y = 9$ $y = 3$	$x = 1, y = 3$
4.8	$5x + 3y = 7$ $2x - y = 5$ $\times 3: \ 6x - 3y = 15$ $\text{add}: \ 11x = 22$ $x = 2$ $5 \times 2 + 3y = 7$ $y = -1$	$x = 2, y = -1$
4.9	$2x + 3y = 9$ $5x - 2y = 13$ $\times 2: \ 4x + 6y = 18$ $\times 3: \ 15x - 6y = 39$ $\text{add}: \ 19x = 57$ $x = 3$ $2 \times 3 + 3y = 9$ $y = 1$	$x = 3, y = 1$
4.10	$3x + 2y = 8$ $5x - 4y = 6$ $\times 4: \ 12x + 8y = 32$ $\times 2: \ 10x - 8y = 12$ $\text{add}: \ 22x = 44$ $x = 2$ $3 \times 2 + 2y = 8$ $y = 1$	$x = 2, y = 1$

4.11	$4x + y = 6$ $3x - 2y = -1$ $\times 2: \ 8x + 2y = 12$ $\text{add}: \ 11x = 11$ $x = 1$ $4 \times 1 + y = 6$ $y = 2$	$x = 1, y = 2$
4.12	$2x + 5y = 3$ $3x - y = 13$ $\times 5: \ 15x - 5y = 65$ $\text{add}: \ 17x = 68$ $x = 4$ $2 \times 4 + 5y = 3$ $y = -1$	$x = 4, y = -1$
word problems		
4.13	$2w + 3t = 3$ $3w + t = 2.75$ $\times 3: \ 9w + 3t = 8.25$ $\text{subtract}: \ 7w = 5.25$ $w = 0.75$ $2 \times 0.75 + 3t = 3$ $t = 0.5$	**wrap: 0.750 KD** **tea: 0.500 KD**
4.14	$4p + n = 2.25$ $2p + 3n = 1.75$ $\times 3: \ 12p + 3n = 6.75$ $\text{subtract}: \ 10p = 5$ $p = 0.5$ $4 \times 0.5 + n = 2.25$ $n = 0.25$	**pen: 0.500 KD** **notebook: 0.250 KD**

4.15	$3s + 2c = 6$ $2s + 5c = 6.75$ $\times 5:\ 15s + 10c = 30$ $\times 2:\ 4s + 10c = 13.5$ $\text{subtract}:\ 11s = 16.5$ $s = 1.5$ $3 \times 1.5 + 2c = 6$ $c = 0.75$	**sandwich: 1.500 KD** **coffee: 0.750 KD**
4.16	$2c + w = 3$ $c + 3w = 2.75$ $\times 3:\ 6c + 3w = 9$ $\text{subtract}:\ 5c = 6.25$ $c = 1.25$ $2 \times 1.25 + w = 3$ $w = 0.5$	**cake: 1.250 KD** **water: 0.500 KD**
4.17	$3a + 4o = 3$ $2a + o = 1.5$ $\times 4:\ 8a + 4o = 6$ $\text{subtract}:\ 5a = 3$ $a = 0.6$ $3 \times 0.6 + 4o = 3$ $o = 0.3$	**apple: 0.600 KD** **orange: 0.300 KD**
4.18	$5p + 2e = 3.4$ $3p + 4e = 4$ $\times 4:\ 20p + 8e = 13.6$ $\times 2:\ 6p + 8e = 8$ $\text{subtract}:\ 14p = 5.6$ $p = 0.4$ $5 \times 0.4 + 2e = 3.4$ $e = 0.7$	**pencil: 0.400 KD** **eraser: 0.700 KD**